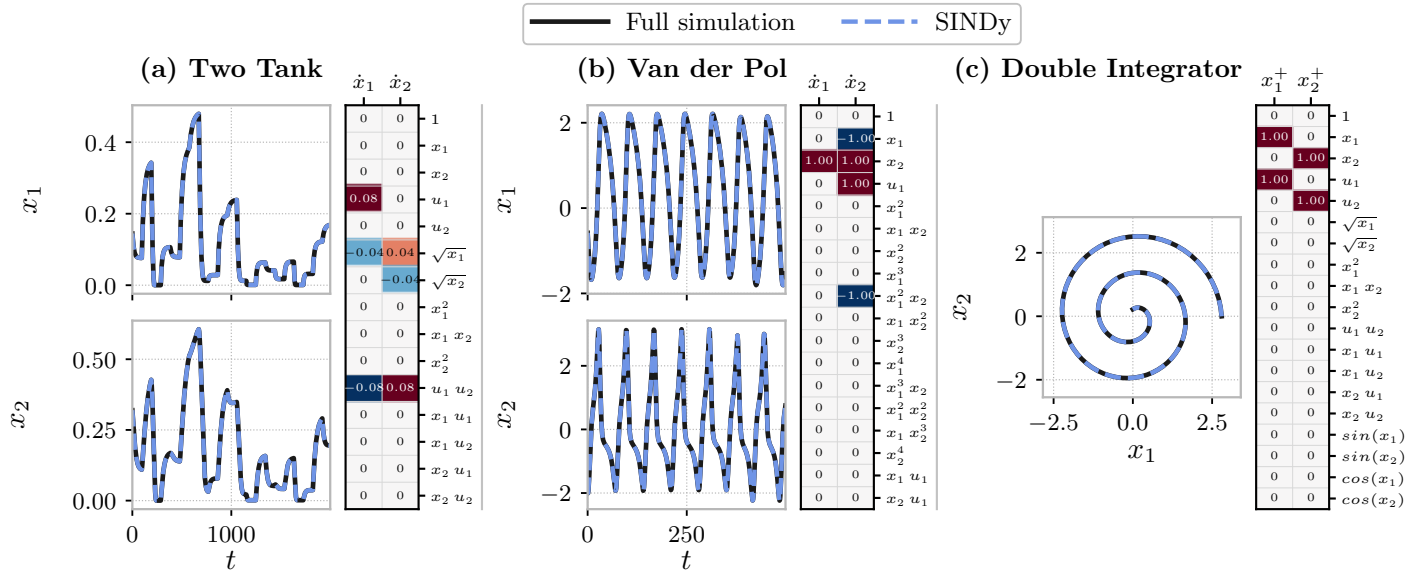

Full simulation
SINDy
(a) Two Tank
(b) Van der Pol
(c) Double Integrator
x_1
x_2
t
$\dot{x}_1$ $\dot{x}_2$
x_1^+ x_2^+
1
x_1
x_2
u_1
u_2
$\sqrt{x_1}$
$\sqrt{x_2}$
x_1^2
$x_1\,x_2$
x_2^2
$u_1\,u_2$
$x_1\,u_1$
$x_1\,u_2$
$x_2\,u_1$
$x_2\,u_2$
x_1^3
$x_1^2\,x_2$
$x_1\,x_2^2$
x_2^3
$x_1^3\,x_2$
$x_1^2\,x_2^2$
$x_1\,x_2^3$
x_2^4
$sin(x_1)$
$sin(x_2)$
$cos(x_1)$
$cos(x_2)$